VALGAN 450
(Valganciclovir Tablets 450 mg)
2081853

Name of the product
VALGAN 450 (Valganciclovir Tablets 450 mg)

Name and strength of Active Ingredient
Each film coated tablet contains 496.3 mg of valganciclovir hydrochloride equivalent to 450 mg of Valganciclovir.

Product Description
Pink, oval, biconvex, film-coated tablets, debossed with 'J' on one side and '156' on the other side

Pharmacodynamics/Pharmacokinetics
Pharmacodynamics:
Mechanism of action:
Valganciclovir is an L-valyl ester (prodrug) of ganciclovir. After oral administration, valganciclovir is rapidly and extensively metabolised to ganciclovir by intestinal and hepatic esterases. It was reported that ganciclovir is a synthetic analogue of 2'-deoxyguanosine and inhibits replication of herpes viruses *in vitro* and *in vivo*. Sensitive human viruses include human cytomegalovirus (HCMV), herpes simplex virus-1 and -2 (HSV-1 and HSV-2), human herpes virus -6, -7 and -8 (HHV-6, HHV-7, HHV8), Epstein-Barr virus (EBV), varicella-zoster virus (VZV) and hepatitis B virus (HBV).

In CMV-infected cells, ganciclovir is initially phosphorylated to ganciclovir monophosphate by the viral protein kinase, pUL97. Further phosphorylation occurs by cellular kinases to produce ganciclovir triphosphate, which is then slowly metabolised intracellularly. Triphosphate metabolism has been shown to occur in HSV- and HCMV- infected cells with half-lives of 18 and between 6 and 24 hours respectively, after the removal of extracellular ganciclovir. As the phosphorylation is largely dependent on the viral kinase, phosphorylation of ganciclovir occurs preferentially in virus-infected cells.

The virustatic activity of ganciclovir is due to inhibition of viral DNA synthesis by: (a) competitive inhibition of incorporation of deoxyguanosine-triphosphate into DNA by viral DNA polymerase, and (b) incorporation of ganciclovir triphosphate into viral DNA causing termination of, or very limited, further viral DNA elongation.

Pharmacokinetics
Absorption
Valganciclovir is a prodrug of ganciclovir. It is well absorbed from the gastrointestinal tract and rapidly and extensively metabolised in the intestinal wall and liver to ganciclovir. Systemic exposure to valganciclovir is transient and low. The absolute bioavailability of ganciclovir from valganciclovir is approximately 60 % across all the patient populations studied and the resultant exposure to ganciclovir is similar to that after its intravenous administration. For comparison, the bioavailability of ganciclovir after administration of 1000 mg oral ganciclovir (as capsules) is 6 - 8 %.

Valganciclovir in HIV positive, CMV positive patients:
The efficacy of ganciclovir in increasing the time-to-progression of CMV retinitis has been reported to correlate with systemic exposure (AUC).

Valganciclovir in solid organ transplant patients:
The systemic exposure of ganciclovir to heart, kidney and liver transplant recipients was similar after oral administration of valganciclovir according to the renal function dosing algorithm.

Food effect:
The inter-individual variation in exposure of ganciclovir decreases when taking Valganciclovir with food. Therefore, it is recommended that Valganciclovir be administered with food.

Distribution:
Because of rapid conversion of valganciclovir to ganciclovir, protein binding of valganciclovir was not determined. Plasma protein binding of ganciclovir was 1 – 2 % over concentrations of 0.5 and 51µg/ml. The steady state volume of distribution (V_d) of ganciclovir after intravenous administration was reported as 0.680 ± 0.161 l/kg (n=114).

Biotransformation
Valganciclovir is rapidly and extensively metabolised to ganciclovir; no other metabolites have been detected. No metabolite of orally administered radiolabelled ganciclovir (1000 mg single dose) accounted for more than 1 – 2 % of the radioactivity recovered in the faeces or urine.

Elimination
Following dosing with Valganciclovir, renal excretion, as ganciclovir, by glomerular filtration and active tubular secretion is the major route of elimination of valganciclovir. Renal clearance accounts was reported for 81.5 % ± 22 % (n=70) of the systemic clearance of ganciclovir. The half-life of ganciclovir from valganciclovir is 4.1 ± 0.9 hours in HIV- and CMV-seropositive patients.

Pharmacokinetics in special population
Patients with renal impairment
Decreasing renal function resulted in decreased clearance of ganciclovir from valganciclovir with a corresponding increase in terminal half-life. Therefore, dosage adjustment is required for renally impaired patients.

Patients undergoing haemodialysis
For patients receiving haemodialysis dose recommendations for Valganciclovir 450 mg film-coated tablets cannot be given. This is because an individual dose of Valganciclovir required for these patients is less than the 450 mg tablet strength. Thus, Valganciclovir film-coated tablets should not be used in these patients.

Patients with hepatic impairment
The safety and efficacy of Valganciclovir film-coated tablets have not been studied in patients with hepatic impairment. Hepatic impairment should not affect the pharmacokinetics of ganciclovir since it is excreted renally and, therefore, no specific dose recommendation is made.

Paediatric population
It was reported that in paediatric solid organ transplant recipients, valganciclovir after administering once daily for 100 days, the pharmacokinetics parameters were similar across organ type and age range and comparable with adults. Population pharmacokinetic modeling suggested that bioavailability was approximately 60%. Clearance was positively influenced by both body surface area and renal function. The once daily dose of Valganciclovir was based on body surface area (BSA) and creatinine clearance (CrCl) derived from a modified Schwartz formula, and was calculated using the equation below:

Paediatric Dose (mg) = 7 x BSA x CrCl (calculated using the modified Schwartz formula) where

$$\text{Mosteller BSA } (m^2) = \sqrt{\frac{\text{Height (cm) x Weight (kg)}}{3600}}$$

$$\text{Schwartz Creatinine Clearance } (ml / min / 1.73m^2) = \frac{k \text{x Height (cm)}}{\text{Serum Creatinine } (mg / dl)}$$

where k = 0.45 for patients aged < 2 years, 0.55 for boys aged 2 to < 13 years and girls aged 2 to 16 years, and 0.7 for boys aged 13 to 16 years.

The dose should not exceed the adult 900 mg dose. In addition, if the calculated Schwartz creatinine clearance exceeds 150 ml/min/1.73m^2, then a maximum value of 150 ml/min/1.73m^2 should be used in the equation. It should be noted that the paediatric dosage algorithm was developed based on pharmacokinetic data only and has not been verified in any published efficacy and safety studies.

Indications
Valganciclovir is indicated:
- for the treatment of cytomegalovirus (CMV) retinitis in acquired immunodeficiency syndrome (AIDS) patients.
- for the prevention of CMV disease in CMV-negative patients who have received a solid organ transplant from a CMV-positive donor.

Recommended dose
Posology
Caution – Strict adherence to dosage recommendations is essential to avoid overdose
Valganciclovir is rapidly and extensively metabolised to ganciclovir after oral dosing.

Standard dosage in adults
Induction treatment of CMV retinitis:
For patients with active CMV retinitis, the recommended dose is 900 mg valganciclovir (two Valganciclovir 450 mg tablets) twice a day for 21 days and, whenever possible, taken with food. Prolonged induction treatment may increase the risk of bone marrow toxicity.

Maintenance treatment of CMV retinitis:
Following induction treatment, or in patients with inactive CMV retinitis, the recommended dose is 900mg valganciclovir (two Valganciclovir 450 mg tablets) once daily and, whenever possible, taken with food. Patients whose retinitis worsens may repeat induction treatment; however, consideration should be given to the possibility of viral drug resistance.

Prevention of CMV disease in solid organ transplantation:
For kidney transplant patients, the recommended dose is 900 mg (two Valganciclovir 450 mg tablets) once daily, starting within 10 days of transplantation and continuing until 100 days post-transplantation. Prophylaxis may be continued until 200 days post-transplantation.

For patients who have received a solid organ transplant other than kidney, the recommended dose is 900 mg (two Valganciclovir 450 mg tablets) once daily, starting within 10 days of transplantation and continuing until 100 days post-transplantation.

Whenever possible, the tablets should be taken with food.

Special dosage instructions
Patients with renal impairment:
Serum creatinine levels or creatinine clearance should be monitored carefully. Dosage adjustment is required according to creatinine clearance, as shown in the table below.

An estimated creatinine clearance (ml/min) can be related to serum creatinine by the following formulae:

$$\text{For males} = \frac{(140 - \text{age [years]}) \text{ x (body weight [kg])}}{(72) \text{ x } (0.011 \text{ x serum creatinine [micromol/l]})}$$

For females = 0.85 × male value

CrCl (ml/min)	Induction dose of valganciclovir	Maintenance/Prevention dose of valganciclovir
≥ 60	900 mg (2 tablets) twice daily	900 mg (2 tablets) once daily
40 – 59	450 mg (1 tablet) twice daily	450 mg (1 tablet) once daily
25 – 39	450 mg (1 tablet) once daily	450 mg (1 tablet) every 2 days
10 – 24	450 mg (1 tablet) every 2 days	450 mg (1 tablet) twice weekly
< 10	not recommended	not recommended

Patients undergoing haemodialysis:
For patients on haemodialysis (CrCl < 10 ml/min) a dose recommendation cannot be given. Thus Valganciclovir film-coated tablets should not be used in these patients.

Patients with hepatic impairment:
Safety and efficacy of Valganciclovir tablets have not been studied in patients with hepatic impairment.

Paediatric population:
The safety and efficacy of Valganciclovir in paediatric patients have not been well-established.

Elderly patients:
Safety and efficacy have not been established in this patient population.

Patients with severe leucopenia, neutropenia, anaemia, thrombocytopenia and pancytopenia: (Please read the warnings and precautions before initiation of therapy)

If there is a significant deterioration of blood cell counts during therapy with Valganciclovir, treatment with haematopoietic growth factors and/or dose interruption should be considered.

Method of administration
Valganciclovir is administered orally, and whenever possible, should be taken with food.

Precautions to be taken before handling or administering the medicinal product

The tablets should not be broken or crushed. Since Valganciclovir is considered a potential teratogen and carcinogen in humans, caution should be observed in handling broken tablets. Avoid direct contact of broken or crushed tablets with skin or mucous membranes. If such contact occurs, wash thoroughly with soap and water, rinse eyes thoroughly with sterile water, or plain water if sterile water is unavailable.

Contraindications
- Valganciclovir is contra-indicated in patients with hypersensitivity to valganciclovir, ganciclovir or to any of the excipients listed.
- Due to the similarity of the chemical structure of Valganciclovir and that of aciclovir and valaciclovir, a cross-hypersensitivity reaction between these drugs is possible. Therefore, Valganciclovir is contra-indicated in patients with hypersensitivity to aciclovir and valaciclovir.
- Valganciclovir is contra-indicated during breast-feeding.

Warnings and precautions
Prior to the initiation of valganciclovir treatment, patients should be advised of the potential risks to the foetus. Ganciclovir was found to be mutagenic, teratogenic, aspermatogenic and carcinogenic, and a suppressor of female fertility in animals. Valganciclovir should, therefore, be considered a potential teratogen and carcinogen in humans with the potential to cause birth defects and cancers. It is also considered likely that Valganciclovir causes temporary or permanent inhibition of spermatogenesis. Women of child bearing potential must be advised to use effective contraception during treatment. Men must be advised to practice barrier contraception during treatment, and for at least 90 days thereafter, unless it is certain that the female partner is not at risk of pregnancy.

Valganciclovir has the potential to cause carcinogenicity and reproductive toxicity in the long term.

Severe leucopenia, neutropenia, anaemia, thrombocytopenia, pancytopenia, bone marrow depression and aplastic anaemia have been observed in patients treated with Valganciclovir (and ganciclovir). Therapy should not be initiated if the absolute neutrophil count is less than 500 cells/µl, or the platelet count is less than 25000/µl, or the haemoglobin level is less than 8 g/dl.

When extending prophylaxis beyond 100 days the possible risk of developing leucopenia and neutropenia should be taken into account.

Valganciclovir should be used with caution in patients with pre-existing haematological cytopenia or a history of drug-related haematological cytopenia and in patients receiving radiotherapy.

It is recommended that complete blood counts and platelet counts be monitored during therapy. Increased haematological monitoring may be warranted in patients with renal impairment. In patients developing severe leucopenia, neutropenia, anaemia and/or thrombocytopenia, it is recommended that treatment with haematopoietic growth factors and/or dose interruption be considered.

The bioavailability of ganciclovir after a single dose of 900 mg valganciclovir is approximately 60 %, compared with approximately 6 % after administration of 1000 mg oral ganciclovir (as capsules). Excessive exposure to ganciclovir may be associated with life-threatening adverse reactions. Therefore, careful adherence to the dose recommendations is advised when instituting therapy, when switching from induction to maintenance therapy and in patients who may switch from oral ganciclovir to valganciclovir as Valganciclovir cannot be substituted for ganciclovir capsules on a one-to-one basis. Patients switching from ganciclovir capsules should be advised of the risk of overdosage if they take more than the prescribed number of Valganciclovir tablets.

In patients with impaired renal function, dosage adjustments based on creatinine clearance are required.

Valganciclovir film-coated tablets should not be used in patients on haemodialysis.

Convulsions have been reported in patients taking imipenem-cilastatin and ganciclovir. Valganciclovir should not be used

size: 240 x 320 mm
Pharma Code No: Front: 1912 & Back: 1913
No of colours: 01 - Black

concomitantly with imipenem-cilastatin unless the potential benefits outweigh the potential risks.

Patients treated with Valganciclovir and (a) didanosine, (b) drugs that are known to be myelosuppressive (e.g. zidovudine), or (c) substances affecting renal function, should be closely monitored for signs of added toxicity.

The use of valganciclovir for the prophylactic treatment of CMV disease in transplantation did not include lung and intestinal transplant patients. Therefore, experience in these transplant patients is limited.

Interaction with other Medicaments

Drug interactions with valganciclovir

Since valganciclovir is extensively and rapidly metabolised to ganciclovir; drug interactions associated with ganciclovir will be expected for valganciclovir.

Effects of other medicinal products on ganciclovir

Imipenem-cilastatin

Convulsions have been reported in patients taking ganciclovir and imipenem-cilastatin concomitantly. These drugs should not be used concomitantly unless the potential benefits outweigh the potential risks.

Probenecid

Probenecid given with oral ganciclovir resulted in statistically significantly decreased renal clearance of ganciclovir (20 %) leading to statistically significantly increased exposure (40 %). These changes were consistent with a mechanism of interaction involving competition for renal tubular secretion. Therefore, patients taking probenecid and Valganciclovir should be closely monitored for ganciclovir toxicity.

Effects of ganciclovir on other medicinal products

Zidovudine

When zidovudine was given in the presence of oral ganciclovir there was a small, but statistically significant increase in the AUC of zidovudine. There was also a trend towards lower ganciclovir concentrations when administered with zidovudine, although this was not statistically significant. However, since both zidovudine and ganciclovir have the potential to cause neutropenia and anaemia, some patients may not tolerate concomitant therapy at full dosage.

Didanosine

Didanosine plasma concentrations were found to be consistently raised when given with ganciclovir (both intravenous and oral). At ganciclovir oral doses of 3 and 6 g/day, an increase in the AUC of didanosine has been observed, and likewise at intravenous doses of 5 and 10 mg/kg/day, an increase in the AUC of didanosine has been observed. There was no clinically significant effect on ganciclovir concentrations. Patients should be closely monitored for didanosine toxicity.

Mycophenolate Mofetil

Since both mycophenolate mofetil (MMF) and ganciclovir have the potential to cause neutropenia and leucopenia, patients should be monitored for additive toxicity.

Stavudine

No clinically significant interactions were observed when stavudine and oral ganciclovir were given in combination.

Trimethoprim

No clinically significant pharmacokinetic interaction was observed when trimethoprim and oral ganciclovir were given in combination. However, there is a potential for toxicity to be enhanced since both drugs are known to be myelosuppressive and therefore both drugs should be used concomitantly only if the potential benefits outweigh the risks.

Other antiretrovirals

At clinically relevant concentrations, there is unlikely to be either a synergistic or antagonistic effect on the activity of either ganciclovir or the other antivirals. Metabolic interactions with, for example, protease inhibitors and non-nucleoside reverse transcriptase inhibitors (NNRTIs) are unlikely due to the lack of P450 involvement in the metabolism of either valganciclovir or ganciclovir.

Other potential drug interactions

Toxicity may be enhanced when valganciclovir is co-administered with, or is given immediately before or after, other drugs that inhibit replication of rapidly dividing cell populations such as occur in the bone marrow, testes and germinal layers of the skin and gastrointestinal mucosa. Examples of these types of drugs are dapsone, pentamidine, flucytosine, vincristine, vinblastine, adriamycin, amphotericin B, trimethoprim/sulpha combinations, nucleoside analogues and hydroxyurea.

Since ganciclovir is excreted through the kidney, toxicity may also be enhanced during co-administration of valganciclovir with drugs that might reduce the renal clearance of ganciclovir and hence increase its exposure. The renal clearance of ganciclovir might be inhibited by two mechanisms: (a) nephrotoxicity, caused by drugs such as cidofovir and foscarnet, and (b) competitive inhibition of active tubular secretion in the kidney by, for example, other nucleoside analogues.

Therefore, all of these drugs should be considered for concomitant use with valganciclovir only if the potential benefits outweigh the potential risks.

Pregnancy and lactation

Pregnancy

There are no data from the use of Valganciclovir in pregnant women. Its active metabolite, ganciclovir, readily diffuses across the human placenta. Based on its pharmacological mechanism of action and reproductive toxicity reported in animals with ganciclovir there is a theoretical risk of teratogenicity in humans.

Valganciclovir should not be used in pregnancy unless the therapeutic benefit for the mother outweighs the potential risk of teratogenic damage to the child.

Breast-feeding

It is unknown if ganciclovir is excreted in breast milk, but the possibility of ganciclovir being excreted in the breast milk and causing serious adverse reactions in the nursing infant cannot be discounted. Therefore, breast-feeding must be discontinued.

Fertility

Women of child-bearing potential must be advised to use effective contraception during treatment. Male patients must be advised to practice barrier contraception during, and for at least 90 days following treatment with Valganciclovir unless it is certain that the female partner is not at risk of pregnancy.

Side effects

Valganciclovir is a prodrug of ganciclovir, which is rapidly and extensively metabolised to ganciclovir after oral administration. The undesirable effects known to be associated with ganciclovir use can be expected to occur with valganciclovir. All of the undesirable effects observed with valganciclovir have been previously observed with ganciclovir. The most commonly reported adverse drug reactions following administration of valganciclovir in adults are neutropenia, anaemia and diarrhoea.

Valganciclovir is associated with a higher risk of diarrhoea compared to intravenous ganciclovir. In addition, valganciclovir is associated with a higher risk of neutropenia and leucopenia compared to oral ganciclovir.

Severe neutropenia (< 500 ANC/μl) is seen more frequently in CMV retinitis patients undergoing treatment with valganciclovir than in solid organ transplant patients receiving valganciclovir.

Adverse reactions reported with either valganciclovir, oral ganciclovir, or intravenous ganciclovir is presented in the table below. The adverse reactions listed were reported in patients with AIDS for the induction or maintenance treatment of CMV retinitis, or in liver, kidney or heart transplant patients for the prophylaxis of CMV disease. The term (severe) in parenthesis in the table indicates that the adverse reaction has been reported in patients at both mild/moderate intensity and severe/life-threatening intensity.

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Body System	Adverse reactions
Infections and infestations	Oral candidiasis, sepsis (bacteraemia, viraemia), cellulitis, urinary tract infection
Blood and lymphatic system disorders	(Severe) neutropenia, anaemia, Severe anaemia, (severe) thrombocytopenia, (severe) leucopenia, (severe) pancytopenia, Bone marrow failure, Aplastic anaemia
Immune system disorders	Anaphylactic reaction
Metabolic and nutrition disorders	Decreased appetite, anorexia
Psychiatric disorders	Depression, anxiety, confusion, abnormal thinking, agitation, psychotic disorder, hallucination
Nervous system disorders	Headache, insomnia, dysgeusia, (taste disturbance), hypoaesthesia, paraesthesia, peripheral neuropathy, dizziness, convulsion, Tremor
Eye disorders	Macular oedema, retinal detachment, vitreous floaters, eye pain, visual disturbance, conjunctivitis
Ear and labyrinth disorders	Ear pain, deafness
Cardiac disorders	Arrhythmia
Vascular disorders	Hypotension
Respiratory, thoracic and mediastinal disorders	Dyspnoea, cough
Gastrointestinal disorders	Diarrhoea, Nausea, vomiting, abdominal pain, upper abdominal pain, dyspepsia, constipation, flatulence, dysphagia, abdominal distension, mouth ulceration, pancreatitis
Hepatobiliary disorders	(Severe) hepatic function abnormal, blood alkaline phosphatase increased, aspartate aminotransferase increased, alanine aminotransferase increased
Skin and subcutaneous disorders	Dermatitis, night sweats, pruritus, alopecia, urticaria, dry skin
Musculoskeletal, connective tissue and bone disorders	Back pain, myalgia, arthralgia, muscle spasms
Renal and urinary disorder	Creatinine clearance renal decreased, renal impairment, haematuria, renal failure
Reproductive system and breast disorders	Male infertility
General disorders and administration site conditions	Fatigue, pyrexia, chills, pain, chest pain, malaise, asthenia
Investigations	Weight decreased, blood creatinine increased

Severe thrombocytopenia may be associated with potentially life-threatening bleeding.

Paediatric population

The most frequently reported adverse reactions on treatment in paediatrics were diarrhoea, nausea, neutropenia, leucopenia and anaemia.

Symptoms and treatment of overdose

Overdose experience with Valganciclovir

Fatal bone marrow depression (medullary aplasia) was reported in one patient, after several days of dosing that was at least 10-fold greater than recommended for the patient's degree of renal impairment (decreased creatinine clearance).

It is expected that an overdose of valganciclovir could also possibly result in increased renal toxicity.

Haemodialysis and hydration may be of benefit in reducing blood plasma levels in patients who receive an overdose of valganciclovir.

Storage condition

Store below 30°C and protect from moisture.

Shelf life

3 Years

Dosage forms and packaging available

60's Tablets; 10 x 10's tablets

Manufacturer

Hetero Labs Limited

Unit V, Sy. No 439,440,441 & 458

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Jadcherla Mandal, Mahaboobnagar Dist.

Telangana, India

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